Big Data and Machine Learning for Applied Economics

Week 04: Classification: From Probability to Decision

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October 12, 2026

Previously...

- **The judge:** cross-validation estimates the out-of-sample error without touching the test set. Judged that way, the champion of the 511 fell to 73rd place and the one-standard-error rule picked the line.
- **The knob:** the penalty replaces the per-variable switch with a continuous price. Ridge divides, the Lasso subtracts and switches off; the judge picks λ.
- **Country-wide:** the 37 census variables leave the error at 19.8; the 33 department dummies bring it down to 15.6. The knob no longer paid: the judge pushed λ to zero and the penalized models landed on OLS.
- **Two judges, two questions:** validating at random answers "can I fill in the map?"; holding out whole municipalities, "can I predict a new municipality?". It costs the k-NN 3.0 points.

What we bring

A model that predicts the Petro vote with a test RMSE of 15.6 points —census + department dummies—, and a discipline: choose with the judge and open the test set exactly once.

Today's question: does Petro win the station, and how sure are we?

The same election, the same stations, the same census; the y changes

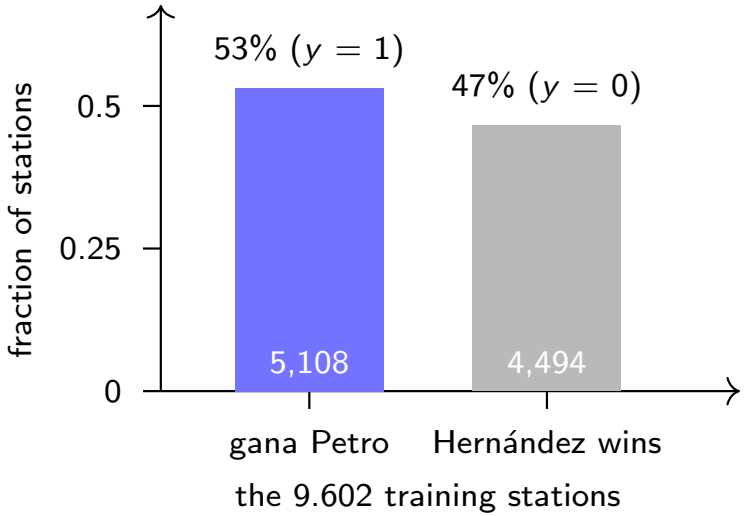
- **What we have:** a number. The census + department regression predicts the vote with a test RMSE of 15.6 points.
- **What the campaign needs:** a decision per station. Which ones it takes for granted (and sends no resources) and which ones it works. It does not care whether Petro gets 60 or 75: it cares **whether he wins**, and **how sure** that is.
- **Today's y:** `gana_petro` $= 1$ if Petro got more votes than Hernández at the station (that is what winning means, not clearing 50 %). Its distribution, before any model: the next slide.

The takeaway

To classify is to estimate a **probability** and then **decide**: the model ranks the stations and the costs of each error set the cutoff.

The y, before any model

As in week 2: describe the target, and pull the first benchmarks from it



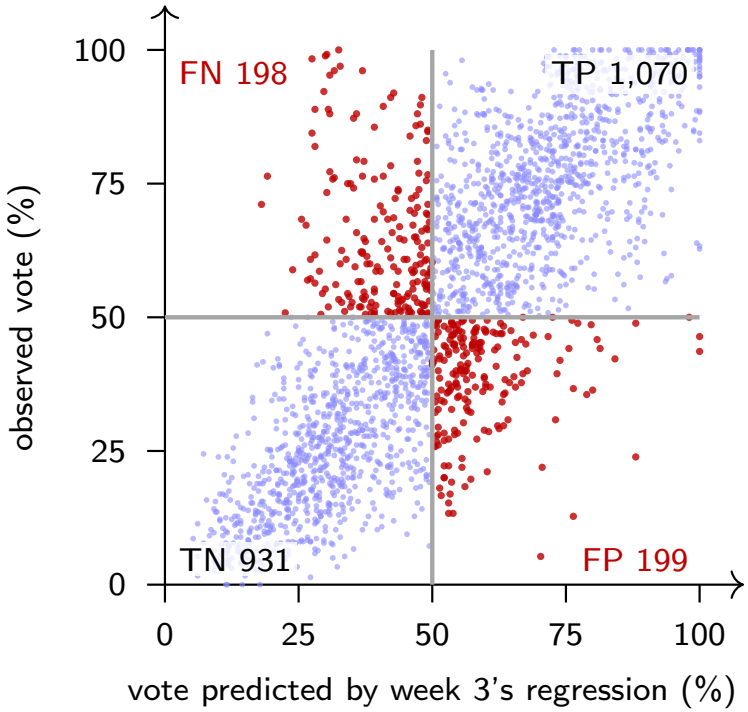
- **The y is a 0 or a 1:** `gana_petro` $= 1$ if Petro got more votes than Hernández at the station, 0 otherwise. In the training set he wins at 53 % (5,108 of 9,602): the classes are nearly balanced.
- **Benchmark 1, with no data at all:** declare "he wins everywhere". On the test set it gets 52.9 % right: the **prevalence**, the fraction of stations with $y = 1$.
- **Benchmark 2, with no model:** the campaign's standing rule, "he wins where he won in 2018". It gets 84.3 % right on the 2,014 stations with a history.

Today's bar

Every model has to beat both benchmarks; and when one class dominates, prevalence inflates *accuracy* (we will see it with 2010).

The shortcut: week 3's regression already classifies

"He wins wherever his predicted vote is higher": the four quadrants are the confusion matrix



- **The shortcut:** nothing new to estimate: week 3's regression predicts the two-candidate vote; "win" if it reaches 50 — **more votes than Hernández**.
- It gets **83.4%** right (benchmark 1: 52.9%); the 2018 rule holds: 84.3% vs. 83.9%.
- In red, the errors against `gana_petro`: the close calls.
- But it delivers **vote points, not a probability**: a 51 and a 95 are the same "win".

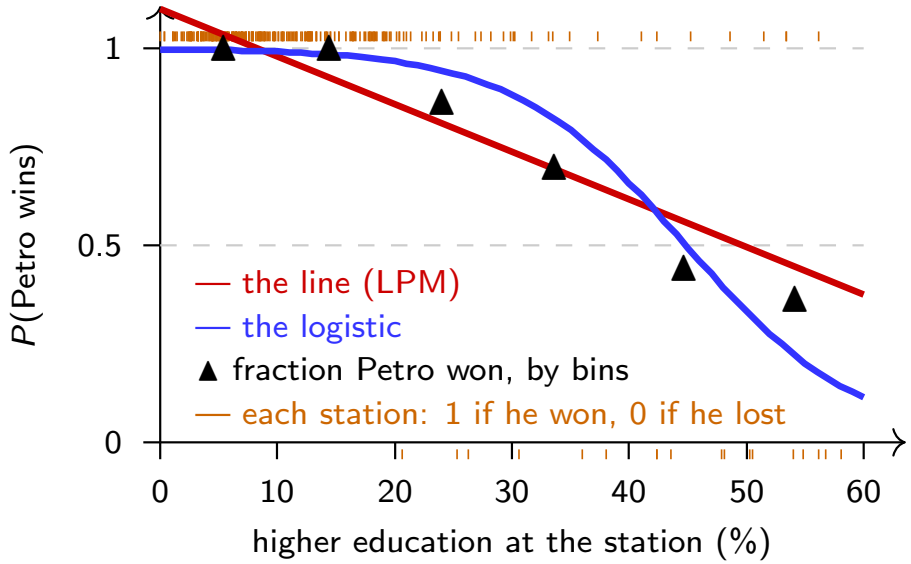
What is missing

A model for $P(\text{win} \mid x)$: with a probability, a formula gives the cheapest cut; with vote points, you grope for it.

1. The probability: from the line to the curve
2. The decision and its metrics
3. Evaluating and comparing: rank, decide, and quantify
4. Problem Set 1
5. Closing

Why not a line? The curve is in the data

Week 2's metro area: 257 stations, one predictor (ISL, section 4.2)



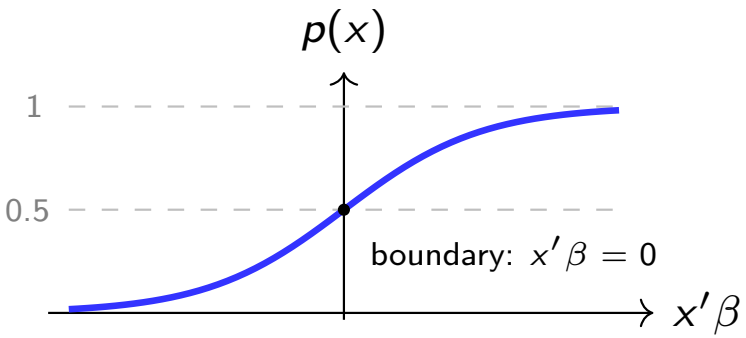
- With $y \in \{0, 1\}$, $\mathbb{E}[y \mid x] = P(y = 1 \mid x)$: the missing $P(\text{win} \mid x)$ can be estimated by OLS on the dummy — the **linear probability model** (LPM): $\hat{p} = 1.10 - 0.0121 \times \text{educ}$. But it leaves $[0, 1]$ at 112 of the 257 stations.
- The triangles do not follow a line: a plateau near 1 and then a drop. For an average marginal effect the LPM works; for **deciding with probabilities**, it does not: the line must be boxed into $[0, 1]$ — the S of the next slide.

Country-wide the same happens: with census + department dummies, the LPM leaves $[0, 1]$ at 371 of the 2,398 test stations (from -0.28 to 1.36).

The logistic: an S for the probability

The same old line, converted into a probability (ISL, section 4.3.1)

$$p(x) = \frac{e^{x'\beta}}{1 + e^{x'\beta}}$$

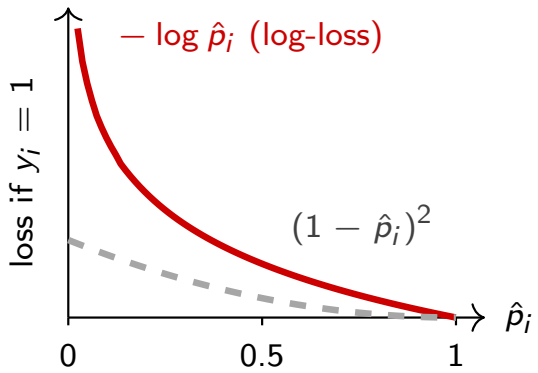


- **The economist's reading:** a **latent advantage** for Petro at the station, $y^* = x'\beta + \varepsilon$; we observe $y = 1$ if $y^* > 0$. Its mean, the **expected advantage** $x'\beta$, is the same old line; the S converts it into a probability.
- With logistic ε you get this S (the *logit*); with normal ε, the *probit*: twins in practice.
- **In reverse:** $\log \frac{p}{1-p} = x'\beta$: the *odds* $p/(1-p)$ in logs are a line after all; that is how the β's will be read (two slides ahead).
- **On the 257 metro-area stations:** expected advantage $= 6.09 - 0.136 \times$ educ; it crosses 0 — the **boundary**, $\hat{p} = 0.5$ — at **44.8 %** higher education: below that, Petro.
- **Parameters:** β. **Hyperparameters:** none.

The likelihood: the RSS of probabilities

Choose the β's that make what we observed most likely (ISL, section 4.3.2)

$$\hat{\beta} \;=\; \arg\max_{\beta}\; \ell(\beta), \qquad \ell(\beta) = \sum_{i=1}^{n} \Big[\, y_i \log p_i + (1-y_i)\log(1-p_i) \,\Big], \qquad p_i = p(x_i).$$



- **The loss:** $-\ell(\beta)/n$ is the **log-loss**. If Petro won and the model gave him $\hat{p} = 0.01$, it pays $-\log 0.01 = 4.6$; the square, at most 1: it punishes **misplaced confidence**.
- **The first-order condition**, $\sum_i x_i(y_i - \hat{p}_i) = 0$, is OLS's with a different prediction: the residual is orthogonal to the x's and, with an intercept, $\sum_i \hat{p}_i = \sum_i y_i$. No closed form: Newton (appendix A1).

What to retain

The logistic is the OLS of probabilities, and its loss is also a metric the judge measures out of sample.

How to read the logistic

Minus 0.136 *what*? Only the sign reads for free; the magnitude must be converted

- **The problem:** in week 2, $\hat{\beta}$ was in vote points (-0.75 per point of education). Today $\hat{\beta}_{\text{educ}} = -0.136$ is in units of the **latent advantage**, and that scale is a convention: **only the sign reads directly —** more education, less Petro.
- **Conversion 1, to odds:** e^{β_j} multiplies the odds $p/(1-p)$: $e^{-0.136} = 0.87$ per point of education; 0.26 per ten points (the *odds ratio* of the papers).
- **Conversion 2, to probability:** $\partial p/\partial x_j = \beta_j\, p\,(1-p)$ depends on the station: -3.4 probability points at the boundary ($p = 0.5$) and -0.1 at a safe one ($p = 0.99$): **it moves the doubtful, not the safe**.

What not to read into the logistic

Two warnings — and the second one asks for the knob again

- **Significant is not predictive:** country-wide, the logistic with week 2's nine variables has 7 coefficients with $p < 0.01$ and gets 68.7% right; "he wins where he won in 2018", with not a single coefficient, 84.3%.
- **Separation:** in Casanare Petro won 3% of the stations and in Putumayo, 98%. With one slope per department, the likelihood pushes some β's toward $\pm\infty$: once again a knob is needed.

We read $\hat{\beta}$ to understand the model, not to estimate effects; opening models without coefficients is session 6.

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From the model to the decision: the cutoff τ

The logistic delivers a probability; the class is ours to assign

- **This part's model:** part 1's logistic, now with week 3's variables — the 37 census variables + the department dummies —, estimated on the 9,602 training stations.
- **What it delivers** for each test station is $\hat{p} = \widehat{P}(\text{win} \mid x) \in [0, 1]$: a probability, **not** a "win/lose".
- **The class comes from a rule of ours:** pick a cutoff τ and declare "win" if $\hat{p} \geq \tau$. The first natural choice: $\tau = 0.5$, declare whichever is more likely.
- **Three objects, not one:** the shortcut's 50 was predicted **vote points**; $\hat{p}$ is a **probability** of winning; τ is the **cut** that turns that probability into a class.

This part's question

What does $\tau = 0.5$ produce? What does the model look like at *all* cutoffs at once? And which τ pays? That is this part's route.

The confusion matrix: the rule's four outcomes

The census + dummies logistic with $\tau = 0.5$, on the 2,398 test stations (ISL, section 4.4.2)

The rule: $\hat{p} \geq \tau = 0.5$

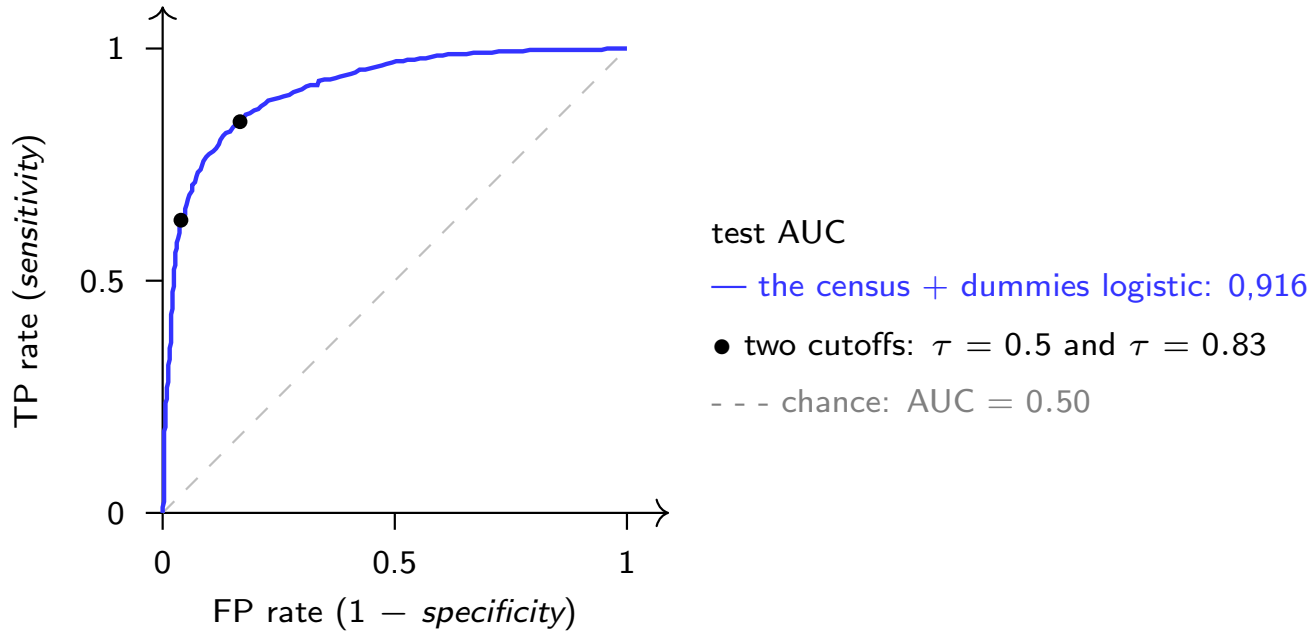
	Pred. win	Pred. lose
Won (1,268)	TP = 1,067	FN = 201
Lost (1,130)	FP = 188	TN = 942

- **Accuracy** $= (TP + TN)/n = 0.84$: the hit rate we have been comparing the benchmarks with.
- **Sensitivity** — in the papers, *recall*; **they are the same metric** — $= TP/(TP + FN) = 0.84$: of the ones he won, how many I found.
- **Specificity** $= TN/(TN + FP) = 0.83$: of the ones he lost, how many I recognized.
- **Precision** $= TP/(TP + FP) = 0.85$: of the ones I called wins, how many really were.

Four readings of the same table, all at $\tau = 0.5$. And at every other cutoff? The next slide.

The ROC curve: all cutoffs at once

Every point on the curve is a confusion matrix (ISL, section 4.4.2, Fig. 4.8)



- **The axes are the metrics we just defined:** vertical, the TP rate (*sensitivity*); horizontal, the FP rate (1− *specificity*). A rule $\hat{p} \geq \tau$ is **one point**; sweeping τ from 1 to 0 draws **the curve**.
- The two black dots: $\tau = 0.5$, the rule from the previous matrix, and a more demanding cut, $\tau = 0.83$, which in a few slides will get a name of its own. And the "AUC" in the legend? The next slide.

The AUC: how well the model ranks, with no cutoff fixed

The area under the ROC curve summarizes every cutoff in a single number

$$\text{AUC} \;=\; P\big(\hat{p}_i > \hat{p}_j \;\big|\; y_i = 1,\; y_j = 0\big)$$

- **The exact reading:** the probability that a station Petro won receives a higher probability than one he lost (appendix A4). 0.5 is chance; 1, perfect ordering. The census + dummies logistic: **0.916**.
- **It does not depend on τ:** it measures the **ranking**, not the decision. That is why it serves to compare models before committing to a cutoff.
- **Ranking is not deciding:** the 2018 vote as a score has an AUC of 0.95, but with its cut at 50 it gets 84.3% right: Petro grew, and of the 241 stations where he got between 40 and 50 in 2018, he won 65%.

And if we move τ? The whole matrix moves

The same $\hat{p}$'s, another cut: nothing is re-estimated

Cutoff 0.5

	Pred. win	Pred. lose
Won	TP = 1,067	FN = 201
Lost	FP = 188	TN = 942

Cutoff 0.83

	Pred. win	Pred. lose
Won	TP = 798	FN = 470
Lost	FP = 45	TN = 1,085

- Raising the cut demands more confidence to declare "win": FPs fall from 188 to 45 and FNs rise from 201 to 470.
- The metrics move in opposite directions: *precision* 0.85 $\rightarrow$ 0.95; *sensitivity* 0.84 $\rightarrow$ 0.63; *accuracy* 83.8% $\rightarrow$ 78.5%.
- **Which τ is the right one?** The mechanics do not say, and neither does the AUC: it is a question about **costs**. The next slide.

The cutoff is an economic decision

Statistics estimates p; economics supplies the costs

The two errors do not cost the same: an **FP** is taking a station for granted and losing it; an **FN**, working one he was going to win anyway. Let c_{FP} and c_{FN} be those costs and p the probability of winning.

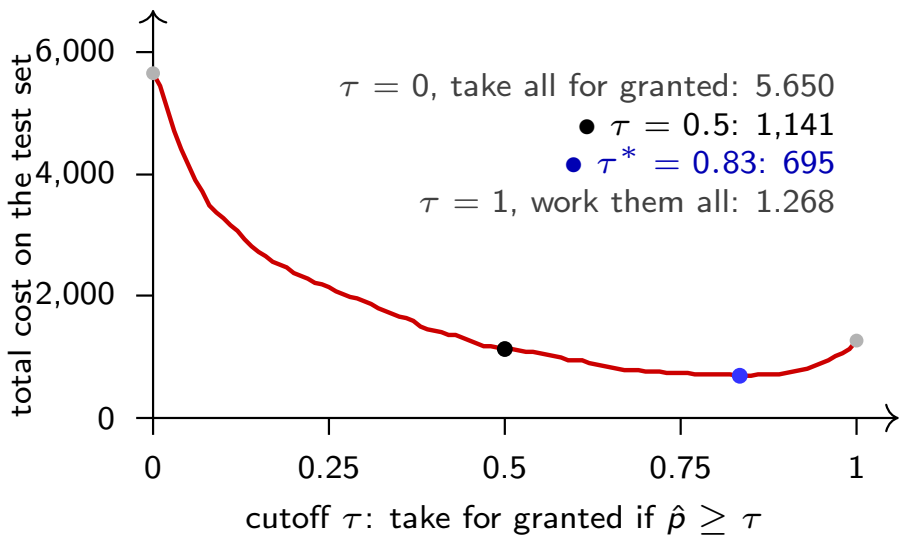
1. **Expected cost of taking it for granted:** $(1-p)\,c_{FP}$ (if he loses it)
2. **Expected cost of working it:** $p\,c_{FN}$ (if he was winning anyway)
3. **Take it for granted when** $(1-p)\,c_{FP} \leq p\,c_{FN}$, that is:

$$\boxed{p \geq \tau^* = \frac{c_{FP}}{c_{FP} + c_{FN}}}$$

- Only the **ratio** of costs matters. If both errors cost the same, $\tau^* = 0.5$: the Bayes rule.
- **The campaign:** getting overconfident and losing costs five times more than working one extra station. $c_{FP} = 5$, $c_{FN} = 1 \Rightarrow \tau^* = 5/6 \approx \mathbf{0.83}$: only the *very* safe is taken for granted.
- The packages' default 0.5 is an implicit economic decision, and almost never the right one.

The cost curve: the minimum lands where the formula said

The campaign's total cost on the test set, for each cutoff

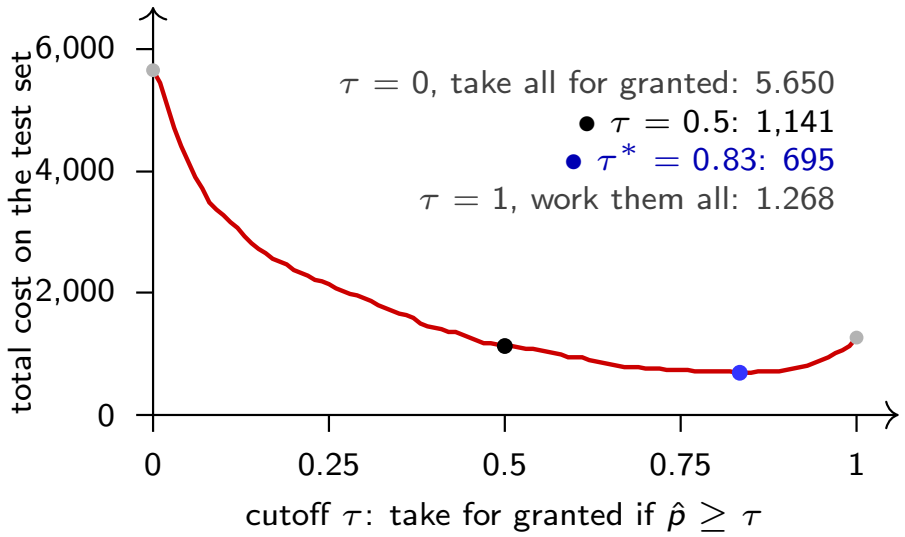


- The minimum lands at $\tau = 0.84$; the formula said 0.83: **the 0.83 from the mechanics was τ^*.**
- The two matrices, with their cost: $\tau = 0.5$ pays **1,141**; τ^*, **695** — and *accuracy* gets worse (83.8 → 78.5%): **it counts errors, it does not weigh costs.**
- With no model: all taken for granted costs 5,650; all worked, 1,268. **Better $\hat{p}$'s would be worth real money**: part 3 measures whether a competitor has them.

Question for the room: Hernández's campaign uses the same probabilities to decide which stations to write off, with the same 5 to 1. Which cutoff does it use?

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Question for the room: Hernández's campaign uses the same probabilities to decide which stations to write off, with the same 5 to 1. Which cutoff does it use? Its expensive error is the opposite one: writing off a station Petro ends up winning. $\tau^* = 1/6 \approx 0.17$ on the same $\hat{p}$. Same model, different economics, different cutoff.

Special case: when one class dominates

Off the main thread, to take home: 2010, where Santos won 88 % of the stations

The y: in 2010, 1 if **Santos** gets more votes than Mockus at the station; in 2022, if **Petro** gets more than Hernández. The same logistic (census blocks and departments), estimated on *its own* training set; the class, with $\hat{p} \geq \tau = 0.5$: from that matrix, *accuracy* and F1 (the harmonic mean of *precision* and *recall*). The AUC uses no τ.

	2010 (Santos–Mockus)	2022 (Petro–Hernández)
Prevalence ($y = 1$)	0.882	0.529
Accuracy	0.901	0.832
F1	0.945	0.840
Balanced accuracy	0.655	0.832
AUC	0.878	0.906

Question for the room: which one is more predictable?

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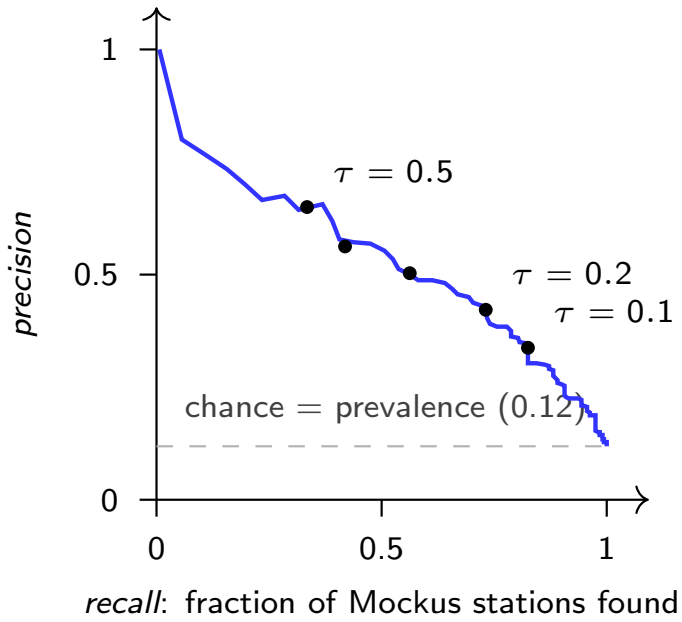
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Question for the room: which one is more predictable? By *accuracy* and F1, 2010; by *balanced accuracy* and AUC, 2022. In 2010 Santos won 88.2%: "Santos wins everywhere" errs on 11.8% and the model on 9.9%. *Accuracy* rewards dominance, not skill: of Mockus's stations, the model recognizes one in three (*specificity* of 0.33).

- To compare elections, regions, or years, the headline metric **cannot depend on prevalence**: AUC or *balanced accuracy* = (*sensitivity* + *specificity*)/2.

Special case (II): rare positives — *precision* and *recall*

Hunting for Mockus's stations in 2010: 11.8 % of the test set



- At $\tau = 0.5$ the model **flags** 109 stations as Mockus's: it finds 33% (*recall*) and is right on 65% of the flagged (*precision*).
- To find 83% you must lower τ to 0.1: it flags 519, and only 34% are truly Mockus's.
- The false-positive rate barely moves from 0.02 to 0.22: it divides by the negatives, which are plentiful, and the ROC barely notices. The PR curve uses no TNs.

Practical rule (Saito and Rehmsmeier, 2015)

Rare positives (Problem Set 1's VIS housing, a fraud, a default) ⇒ report ROC **and** PR, with the prevalence next to them. With balanced classes, like the 2022 stations, the ROC is enough.

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The competitor: the country-wide LPM

Week 2's squared loss on the 0/1 y, with census + dummies and the 10 folds

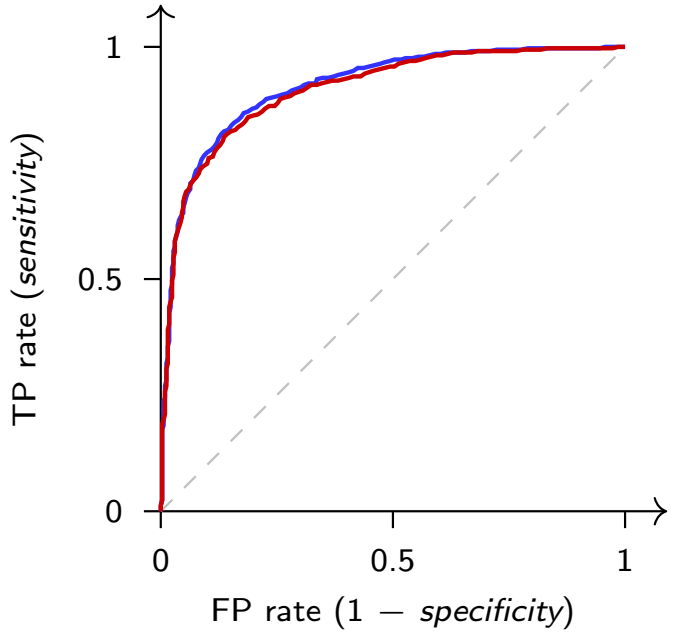
$$\hat{\beta}^{\text{LPM}} = \arg\min_{\beta} \frac{1}{n} \sum_{i} \left(y_i - x_i'\beta\right)^2, \qquad y_i \in \{0, 1\}.$$

- **It is exactly week 3's regression; only the y changed.** The same 70 columns, the same OLS; each $\hat{\beta}_j$ reads in probability points.
- **It ranks almost as well as the logistic:** test AUC of 0.909 against 0.916 (and 0.916 against 0.920 in validation).
- **But its "probabilities" are not probabilities:** on the test set it leaves $[0, 1]$ at 371 stations (running from -0.28 to 1.36): the country-wide version of the metro area's 112 of 257. For ranking it works; for the cost cutoff, it does not.

The two full curves, one on top of the other: the next slide.

The two curves, the same test set

The logistic's ROC and the LPM's, on the same 2,398 stations



test AUC

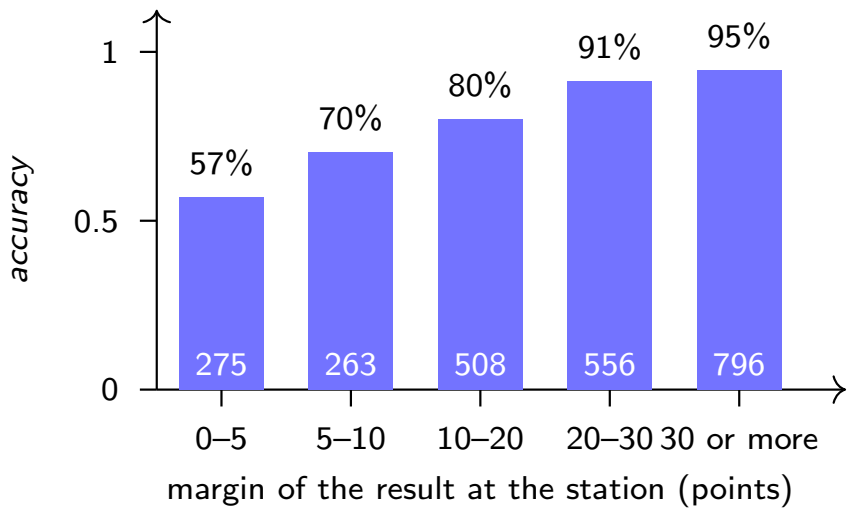
— logistic, census + dummies: 0,916

— LPM, census + dummies: 0,909

- **Ranking is not everything:** the LPM ranks almost the same (0.909 against 0.916), but 371 of its $\hat{p}$'s are not probabilities: the cost rule cannot use it.
- **Today's best:** the census + dummies logistic — it ranks a bit better *and* delivers true probabilities, which is what τ^* demands.

Where it fails: the close-call stations

Accuracy of the census + dummies logistic ($\tau = 0.5$), by the station's margin



- At the 275 stations decided by fewer than 5 points it gets 57% right; above 30, 95%.
- It is week 1's irreducible error, in classification form: where the census cannot tell winning from losing by one vote, no model will.
- In rural areas it does slightly worse: 82.6% against 85.4% in town centers.

Today's table: the benchmarks and the two models

The same 2,398 test stations, opened once; week 3's same 70 columns

Model	CV AUC	Test AUC	*Accuracy* (0.5)
Benchmark 1: "wins everywhere"	—	0.50 (chance)	52.9%
Benchmark 2: the 2018 rule†	—	0.95	84.3%
LPM, census + dept. dummies	0.916	0.909	82.9%
Logistic, census + dept. dummies	0.920	**0.916**	**83.8%**

† On its 2,014 stations with a history; its AUC is that of the 2018 vote as a score.

- **Do they beat the benchmarks?** Benchmark 1, by far. The 2018 rule holds up (84.3% against the shortcut's 83.9% on those stations): today's census barely ties yesterday's history.
- **Who ranks better?** The logistic (0.916), with the LPM on its heels — but with no probabilities for the cost rule.
- **And when deciding?** With 5-to-1 costs the cutoff is $\tau^* = 0.83$, which demands true $\hat{p}$'s: the S. The same model went from paying 1,141 to paying 695.

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Problem Set 1 is released today

Another problem, the same tools. Due: Tuesday, October 20, 8:00 a.m.

A single dataset: apartment and house sale listings in Cali, with a dictionary and a fixed split. In the test listings the price and the VIS status are hidden.

Part	Session	Pts	What you do
1. Benchmark and regression	2	15	the mean and OLS; RMSE, MAE, and R^2; predicted vs. observed
2. Validation and regularization	3	25	random vs. by-zone CV; preprocessing inside the fold; Ridge, Lasso, EN; k-NN
3. Classification	4	15	is it VIS?: benchmark and logistic; matrix and metrics; ROC and AUC; PR (the rare class); cost-based cutoff
4. Final model	2–4	15	choose with validation, refit, and predict the test set once
Course ranking		20	test RMSE of the price; first place 20, last 10
Reproducibility		10	`00_run.R` runs start to finish

In pairs; the session 2–4 practice scripts are the starting point. **Debrief** with the ranking in session 7.

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Recap

1. **First the y and its benchmarks:** the prevalence and the standing rule; every model has to beat them.
2. **To classify is to estimate a probability and then decide:** from the LPM to the logistic (the S, the likelihood, and the log-loss, which punishes misplaced confidence).
3. **The confusion matrix and its metrics:** *accuracy*, *precision*, *sensitivity* (= *recall*), and *specificity*; with unbalanced classes *accuracy* deceives (2010), and with rare positives you look at *precision–recall*.
4. **The cutoff is economics:** $\tau^* = c_{FP}/(c_{FP} + c_{FN})$; with 5 to 1, the cost fell from 1,141 to 695 with the same model, even as *accuracy* got worse. The AUC ranks with no cutoff needed.
5. **Two models, one lesson:** they rank almost the same (0.916 and 0.909); the LPM delivers no probabilities (371 outside $[0, 1]$) and the logistic does, which is what the cost rule asks for. And the 2018 rule remains a serious bar.

Getting ready for next class...

To get the most out of next session

This week's readings:

- ISL, ch. 8 (trees, bagging, random forests, and boosting).
  Guide: figures 8.1 to 8.3 (how a tree splits the space) and 8.8 (forests vs. bagging) are the ones from class; section 8.2.3 is boosting.

Problem Set 1: released today; due Tuesday, October 20 at 8:00 a.m., before session 6.

Next session — trees, forests, and boosting: until today every structural model was linear in $x'\beta$, and the department was only allowed to move the level. A tree splits the territory into groups of similar stations *on its own*, and predicts a constant in each; a forest averages hundreds of trees over bootstrap resamples; boosting fits trees to the residuals. They work for both of the case's questions, `voto_petro` and `gana_petro`.

R application: the same stations and the case's two questions: a tree you can read, a forest, and a boosting model, against today's and week 3's best, on the same test set.

Essential references

- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2. ed.). Springer. Secs. 2.2.3, 4.1–4.5. [ISL]
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning* (2. ed.). Springer. Ch. 4. [ESL]
- Fawcett, T. (2006). An Introduction to ROC Analysis. *Pattern Recognition Letters*, 27(8), 861–874.
- Saito, T., & Rehmsmeier, M. (2015). The Precision-Recall Plot Is More Informative than the ROC Plot on Imbalanced Datasets. *PLOS ONE*, 10(3), e0118432.
- Murphy, A. H. (1973). A New Vector Partition of the Probability Score. *Journal of Applied Meteorology*, 12(4), 595–600. [appendix]
- Chawla, N., Bowyer, K., Hall, L., & Kegelmeyer, W. (2002). SMOTE: Synthetic Minority Over-sampling Technique. *JAIR*, 16, 321–357. [appendix]
- Cortes, C., & Vapnik, V. (1995). Support-Vector Networks. *Machine Learning*, 20(3), 273–297. [appendix]
- Gelvez, J. D., Cardiles, A., Martínez-González, E. F., & Muñoz, M. (2026). How Predictable Is an Election? Working paper. [The source of the case data.]

6 Appendix: going deeper

A1. The logistic likelihood: gradient and Newton

1. The log-likelihood, using $1 - p_i = 1/(1 + e^{x_i'\beta})$:

$$\ell(\beta) \;=\; \sum_{i=1}^{n} \Big[y_i \log p_i + (1 - y_i) \log(1 - p_i) \Big] \;=\; \sum_{i=1}^{n} \Big[y_i \, x_i'\beta - \log\left(1 + e^{x_i'\beta}\right) \Big].$$

2. Gradient and Hessian (with $\partial \log(1 + e^{x_i'\beta})/\partial\beta = x_i \, p_i$):

$$\frac{\partial \ell}{\partial \beta} \;=\; \sum_{i=1}^{n} x_i \big(y_i - p_i\big), \qquad \frac{\partial^2 \ell}{\partial \beta \, \partial \beta'} \;=\; -\sum_{i=1}^{n} p_i(1 - p_i) \, x_i x_i' \;\equiv\; -\mathbf{X}'\mathbf{W}\mathbf{X}.$$

3. No closed form (p_i depends on β nonlinearly). Newton–Raphson iterates

$$\beta^{(t+1)} \;=\; \beta^{(t)} + (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'(\mathbf{y} - \mathbf{p}),$$

a weighted regression at each step (*IRLS*); the Hessian is negative definite, so ℓ is concave and the maximum is unique if it exists.

4. Reading: with an intercept, $\sum_i \hat{p}_i = \sum_i y_i$. The "residual per regressor" $x_i(y_i - p_i)$ is what logistic boosting fits with trees (session 5).

A2. Generative models: LDA, QDA, and Naive Bayes

ISL's other family (section 4.4): model x within each class and flip with Bayes

$$P(y = k \mid x) = \frac{\pi_k f_k(x)}{\sum_l \pi_l f_l(x)},$$

with $\pi_k = P(y = k)$ and f_k the density of x within class k.

- **LDA:** $f_k = N(\mu_k, \Sigma)$ with the **same** covariance. The discriminant $\delta_k(x) = x'\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k'\Sigma^{-1}\mu_k + \log \pi_k$ is **linear in** x: a linear boundary, like the logistic (ISL 4.5.1 shows both have linear log-odds; they differ in how they estimate).
- **QDA:** one covariance per class $\Rightarrow$ quadratic boundary; $Kp(p+1)/2$ covariance parameters: more flexible, more variance.
- **Naive Bayes:** x's independent within each class, $f_k(x) = \prod_j f_{kj}(x_j)$. False and useful when p is huge and data are scarce (the spam filter).

Why they are not in the main body

With large n and no normality, the logistic is more robust: it does not require assuming how the x's are distributed.

A3. More than two classes, and counts

Multinomial logistic (ISL 4.3.5): with K classes, one $x'\beta_k$ per class and a normalization that allocates the probability,

$$P(y = k \mid x) \;=\; \frac{e^{x'\beta_k}}{\sum_{l=1}^{K} e^{x'\beta_l}} \qquad \text{(la } \textit{softmax}\text{)},$$

with a reference class whose β is fixed at zero. Example: which of the first round's top three wins the station? The same *softmax* is the last layer of a neural network (session 7).

- **Ordered categories** (stratum, education level): ordered logit, or treat the scale as numeric. Today's metrics extend class by class ("one against the rest").
- **Counts** (crimes per beat, claims): Poisson regression, $\log \mathbb{E}[y \mid x] = x'\beta$ (ISL 4.6). The same logic: a link function that respects the support of y.

A4. The AUC as a probability and the Brier decomposition

1. AUC = the probability of ranking a pair correctly. With n_+ positives and n_- negatives, the Mann–Whitney statistic

$$\widehat{\mathrm{AUC}} = \frac{1}{n_+ n_-} \sum_{i \in +} \sum_{j \in -} \left[\mathbb{I}\{\hat{p}_i > \hat{p}_j\} + \tfrac{1}{2}\, \mathbb{I}\{\hat{p}_i = \hat{p}_j\} \right]$$

is the area under the empirical ROC; it depends on neither the cutoff nor the prevalence.

2. The Brier decomposes (Murphy, 1973). With B groups by $\hat{p}$, of size n_b, mean prediction $\bar{p}_b$, and observed frequency $\bar{y}_b$:

$$\underbrace{\tfrac{1}{n}\textstyle\sum_i (\hat{p}_i - y_i)^2}_{\text{Brier}} \approx \underbrace{\tfrac{1}{n}\textstyle\sum_b n_b (\bar{p}_b - \bar{y}_b)^2}_{\text{calibration}} - \underbrace{\tfrac{1}{n}\textstyle\sum_b n_b (\bar{y}_b - \bar{y})^2}_{\text{resolution}} + \underbrace{\bar{y}(1-\bar{y})}_{\text{uncertainty}}.$$

- **Calibration:** the distance to the diagonal in appendix A7's decile figure (lower is better). **Resolution:** how much the probabilities separate the groups (higher is better).
- A high AUC with a bad Brier is possible: good resolution, bad calibration.

A5. Class imbalance: resample or decide better

With 2 % positives (a fraud, a default), the likelihood barely notices the rare class. **The remedies**, from simplest to most elaborate:

1. **Move the cutoff** with the costs: almost always enough if the model ranks well and is calibrated.
2. **Weights in the loss** (*class weights*): make the rare class's errors more expensive, without touching the data.
3. **Undersample** the majority class or **oversample** the minority; **SMOTE** (Chawla et al., 2002) creates synthetic positives between neighbors, $x_{new} = x_i + u\,(x_{nn} - x_i)$, $u \sim U(0,1)$.

Two non-negotiable warnings

(i) All resampling happens **inside each fold and only on the training data** (golden rule #3). (ii) Resampling **distorts** $\hat{p}$: you must recalibrate (appendix A7) before using a cost cutoff.

A6. Support vector machines: the margin intuition

Outside the course's main body; for whoever needs it (ISL, ch. 9)

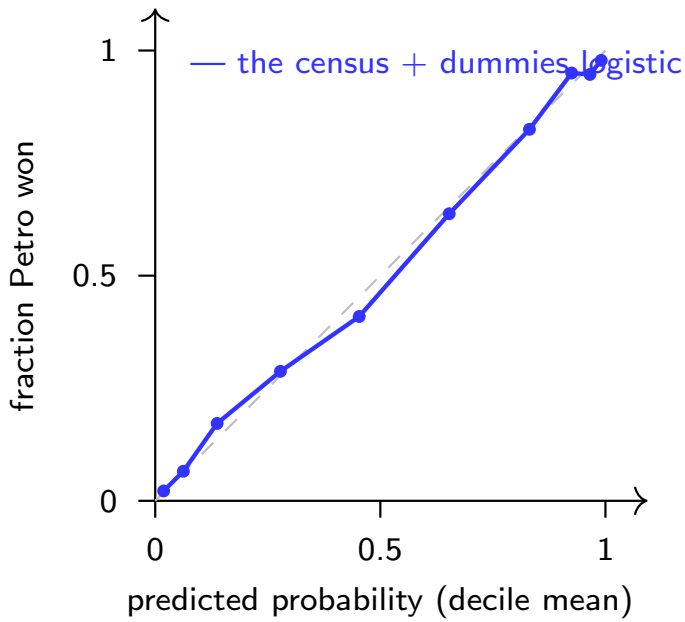
- Another philosophy: find the hyperplane that separates the classes with the **widest margin**; only the points on the margin (the *support vectors*) determine the solution.
- **Soft margin** (C): allow violations at a cost; it is the knob, chosen by CV.
- **Kernel:** replacing inner products with $K(x, x')$ gives nonlinear boundaries without building the space.

Why it is not in the main body

It delivers scores, not probabilities (they must be calibrated), it scales poorly with large n, and in applied economics it rarely shows up next to the penalized logistic and tree ensembles. Reference: Cortes and Vapnik (1995).

A7. Calibration and Brier: do the $\hat{p}$'s tell the truth?

The test set in deciles of $\hat{p}$: the mean probability against the fraction Petro won



- On the diagonal, the model **tells the truth**. The cutoff τ^* takes $\hat{p}$ at face value: without calibration, it decides badly even if the AUC is high.
- **The Brier** $= \frac{1}{n}\sum_i (\hat{p}_i - y_i)^2$ is week 2's MSE applied to probabilities: the logistic, 0.115; the LPM, 0.124; saying the prevalence everywhere, 0.249.
- The LPM does not even make it whole onto this figure: 371 of its $\hat{p}$'s lie outside $[0, 1]$. Ranking costs it nothing; being a probability does.